IGCSE PHYSICS SYLLABUS TRACKING
2025-27

Topic 1 - Motion, Forces & Energy

1.1 **Physical Quantities & Measurement Techniques**
- 1.1.1 ✓Measurement
- 1.1.2 ✓Scalars & Vectors
- 1.1.3 ✓Calculating with Vectors

1.2 **Motion**
- 1.2.1 ✓Speed & Velocity
- 1.2.2 ✓Acceleration
- 1.2.3 ✓Distance-Time Graphs
- 1.2.4 ✓Speed-Time Graphs
- 1.2.5 ✓Calculating Acceleration from Speed-Time Graphs
- 1.2.6 ✓Freefall

1.3 **Mass, Weight & Density**
- 1.3.1 ✓Mass & Weight
- 1.3.2 ✓Density
- 1.3.3 ✓Measuring Density
- 1.3.4 ✓Floating

1.4 **Effects of Forces**
- 1.4.1 ✓Resultant Forces
- 1.4.2 ✓Newton's First Law
- 1.4.3 ✓Newton's Second Law
- 1.4.4 ✓Investigating Force & Extension
- 1.4.5 ✓Hooke's Law
- 1.4.6 ✓Circular Motion
- 1.4.7 ✓Friction

1.5 **Moments**
- 1.5.1 ✓Moments
- 1.5.2 ✓Equilibrium
- 1.5.3 ✓Centre of Gravity
- 1.5.4 ✓Investigating Centre of Gravity

1.6 **Momentum**
- 1.6.1 ✓Momentum
- 1.6.2 ✓Impulse

1.7 **Energy, Work & Power**
- 1.7.1 ✓Energy Stores & Transfers
- 1.7.2 ✓Kinetic Energy
- 1.7.3 ✓Gravitational Potential Energy
- 1.7.4 ✓Conservation of Energy
- 1.7.5 ✓Work Done
- 1.7.6 ✓Power
- 1.7.7 ✓Efficiency

1.8 **Energy Sources**

- 1.8.1 ✓Energy from the Sun
- 1.8.2 ✓Energy from Fuels
- 1.8.3 ✓Energy from Water
- 1.8.4 ✓Geothermal Energy
- 1.8.5 ✓Nuclear Fusion

1.9 **Pressure**

- 1.9.1 ✓Pressure & Forces
- 1.9.2 ✓Pressure in a Liquid

Topic 2 – Thermal Physics

2.1 **Kinetic Particle Model of Matter**

- 2.1.1 ✓States of Matter
- 2.1.2 ✓Molecular Matter
- 2.1.3 ✓Temperature & Pressure
- 2.1.4 ✓Brownian Motion
- 2.1.5 ✓Gases & Absolute Temperature

2.2 **Thermal Properties & Temperature**

- 2.2.1 ✓Thermal Expansion
- 2.2.2 ✓Specific Heat Capacity
- 2.2.3 ✓Investigating Specific Heat Capacity
- 2.2.4 ✓Melting & Boiling
- 2.2.5 ✓Evaporation

2.3 **Transfer of Thermal Energy**

- 2.3.1 ✓Demonstrating Conduction
- 2.3.2 ✓Thermal Conduction
- 2.3.3 ✓Convection
- 2.3.4 ✓Radiation
- 2.3.5 ✓The Greenhouse Effect
- 2.3.6 ✓Investigating IR Radiation
- 2.3.7 ✓Consequences of Thermal Energy Transfer

Topic 3 – Waves

3.1 **General Properties of Waves**

- 3.1.1 ✓Features of Waves
- 3.1.2 ✓The Wave Equation
- 3.1.3 ✓Transverse & Longitudinal Waves
- 3.1.4 ✓Wave Behaviour
- 3.1.5 ✓Ripple Tank

3.2 **Light**

- 3.2.1 ✓Reflection of Light
- 3.2.2 ✓Investigating Reflection
- 3.2.3 ✓Refraction of Light
- 3.2.4 ✓Snell's Law
- 3.2.5 ✓Total Internal Reflection

3.2.6 ✓Thin Lenses
3.2.7 ✓Real & Virtual Images
3.2.8 ✓Uses of Lenses
3.2.9 ✓Dispersion of Light

3.3 **Electromagnetic Spectrum**

3.3.1 ✓Electromagnetic Waves
3.3.2 ✓Uses of Electromagnetic Waves
3.3.3 ✓Dangers of Electromagnetic Waves
3.3.4 ✓Communications
3.3.5 ✓Digital & Analogue Signals

3.4 **Sound**

3.4.1 ✓Sound Waves
3.4.2 ✓Measuring the Speed of Sound
3.4.3 ✓Effects of Sound Waves
3.4.4 ✓Ultrasound

Topic 4 – Electricity and Magnetism

4.1 **Simple Phenomena of Magnetism**

4.1.1 Magnetism
4.1.2 Magnets
4.1.3 Magnetic Fields
4.1.4 Plotting Magnetic Fields

4.2 **Electrical Quantities**

4.2.1 Electric Fields & Charges
4.2.2 Demonstrating Electric Charges
4.2.3 Electric Fields
4.2.4 Investigating Conductors & Insulators
4.2.5 Current
4.2.6 Electromotive Force & Potential Difference
4.2.7 Resistance
4.2.8 Resistance of a Wire
4.2.9 Electrical Energy
4.2.10 Electrical Power

4.3 **Electric Circuits & Electrical Safety**

4.3.1 Circuit Diagrams & Circuit Components
4.3.2 Current in Circuits
4.3.3 EMF & Potential Difference in Circuits
4.3.4 Combined Resistance
4.3.5 Potential Dividers
4.3.6 Electrical Safety

4.4 **Electromagnetic Effects**

4.4.1 Electromagnetic Induction
4.4.2 Demonstrating Induction
4.4.3 The A.C. Generator
4.4.4 Magnetic Effect of a Current

4.4.5 Investigating the Field Around a Wire
4.4.6 Force on a Current-Carrying Conductor
4.4.7 Electric Motors
4.4.8 Transformers
4.4.9 Transformer Calculations

Topic 5 – Nuclear Physics

5.1 **The Nuclear Model of the Atom**
5.1.1 The Atom
5.1.2 The Nucleus
5.1.3 Protons, Neutrons & Electrons
5.1.4 Fission & Fusion
5.2 **Radioactivity**
5.2.1 Background Radiation
5.2.2 Types of Radiation
5.2.3 Ionising Power & Deflection
5.2.4 Radioactive Decay
5.2.5 Half-Life
5.2.6 Uses of Radiation
5.2.7 Dangers of Radiation

Topic 6 – Space Physics

6.1 **Earth & The Solar System**
6.1.1 The Earth, Moon & Sun
6.1.2 Calculating Orbital Speeds
6.1.3 The Solar System
6.1.4 Orbiting Bodies
6.1.5 Gravitational Effects on Orbits
6.2 **Stars & The Universe**
6.2.1 The Sun as a Star
6.2.2 Stars
6.2.3 The Expanding Universe
6.2.4 The Big Bang Theory
6.2.5 Hubble & The Age of the Universe